

PSP Refresh Staging databases - Oracle

- [Refresh Staging Oracle Monolith database](#)
- [Post refresh steps for Monolith Oracle staging](#)
 - [Oracle Range & Oracle Hash Staging database](#)
 - [Oracle Hash Staging](#)
- [Refresh Staging Aurora Postgres Audit database](#)
- [Post refresh steps for Monolith Aurora Postgres staging](#)
- [Rename databases to actual staging db name](#)
- [Sanity Check](#)
- [Cleanup old staging databases](#)

Refresh Staging Oracle Monolith database

- Restore latest prod snapshot to new instance pspparmo-new. Refer existing staging database (pspparmo) for configuration/parameter reference.

RDS > Snapshots > Restore snapshot

Restore snapshot

You are creating a new DB instance or DB cluster from a snapshot. The default VPC security group and parameter group are selected for the new DB instance or DB cluster, but you can change these settings.

DB instance settings

DB engine

Oracle Enterprise Edition ▼

License model

bring-your-own-license ▼

Settings

DB snapshot ID
The identifier for the DB snapshot.
rds:pspuwp01-2023-03-17-05-06

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

pspparmo-new

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Availability & durability

Multi-AZ deployment [Info](#)

- ☐ Create a standby instance (recommended for production usage)
Creates a standby in a different Availability Zone (AZ) to provide data redundancy, eliminate I/O freezes, and minimize latency spikes during system backups.
- ☒ Do not create a standby instance

Connectivity [Info](#)



Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

vpc-2 (vpc-0d8f1534aab54ac6) ▼

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

prod-clusterdb-vpc-2 ▼

Public access [Info](#)

- ☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
- ☒ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing
Choose existing VPC security groups

☐ Create new
Create new VPC security group

Existing VPC security groups

Choose one or more options ▼

parallel-clusterdb-vpc-2 ✕

bastionToDB-BastionToDBSecurityGroup-For-Parallel ✕

Availability Zone [Info](#)

No preference ▼

Certificate authority - optional [Info](#)

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-2019 (default) ▼

If you don't select a certificate authority, RDS chooses one for you.

▼ Additional configuration

Database port [Info](#)

TCP/IP port that the database will use for application connections.

2632

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

- ☐ Standard classes (includes m classes)
- ☒ Memory optimized classes (includes r and x classes)

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

- ☐ Standard classes (includes m classes)
- ☒ Memory optimized classes (includes r and x classes)
- ☐ Burstable classes (includes t classes)

db.r5.16xlarge
64 vCPUs 512 GiB RAM Network: 13,600 Mbps ▼

- ☒ Include additional memory configurations [Info](#)
- ☐ Include previous generation classes

▼ Additional configuration - optional

Processor Features

- ☐ Override default values [Info](#)
You can change the number of CPU cores and threads per core on the DB instance class.

Storage

Storage type [Info](#)

Provisioned IOPS SSD (io1)
Flexibility in provisioning I/O ▼

Allocated storage [Info](#)

47500 GiB

The minimum value is 43,000 GiB and the maximum value is 65,536 GiB

Provisioned IOPS [Info](#)

35000 IOPS

The minimum value is 21,500 IOPS and the maximum value is 60,000 IOPS. The storage size ratio must be between 0.1 and 1000. The storage size ratio is the ratio between allocated storage and the Provisioned IOPS.

⚠ Scaling your storage or IOPS can negatively affect DB instance performance until the storage optimization operation completes. [Learn more](#) [🔗](#)

💡 For a workload with 50% writes and 50% reads running on an m2.4xlarge instance, you can realize up to 25,000 IOPS. However, by provisioning more than this limit, you might be able to achieve lower latency and higher throughput. Your actual realized IOPS might vary from the amount you provisioned based on your database workload and instance type. [Learn more](#) [🔗](#)

Database authentication

Database authentication options [Info](#)

- ☒ Password authentication
Authenticates using database passwords.
- ☐ Password and Kerberos authentication
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

Encryption

Encryption [Info](#)

- ☒ Enable Encryption
Choose to encrypt the copy of the source DB snapshot. Master key IDs and aliases appear in the list after they have been created using KMS. You cannot remove encryption from an encrypted DB snapshot.

AWS KMS key [Info](#)

rds-pspuwp01 ▼

Account
893547637742

KMS key ID
d8e5eb49-fe55-44e4-9fb4-7911b9fc1328

▼ Additional configuration

Database options, backup turned on, backtrack turned off, CloudWatch Logs, maintenance, delete protection turned on

Database options

Initial database name [Info](#)

PSPPARMO

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group [Info](#)

oracle-ee-19-psp-prod-clusterdb-vpc-2 ▼

Option group [Info](#)

oracle-ee-19-prod-clusterdb-vpc-2 ▼

Backup

☒ Copy tags to snapshots

Log exports

Select the log types to publish to Amazon CloudWatch Logs

- ☒ Alert log
☐ Audit log
☐ Listener log
☐ Oracle Management Agent log
☐ Trace log

IAM role

The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS service-linked role

 Ensure that general, slow query, and audit logs are turned on. Error logs are enabled by default. [Learn more](#)

Maintenance

Auto minor version upgrade [Info](#)

- ☐ **Enable auto minor version upgrade**
Enabling auto minor version upgrade will automatically upgrade to new minor versions as they are released. The automatic upgrades occur during the maintenance window for the database.

Deletion protection

- ☒ **Enable deletion protection**
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

Cancel

Restore DB instance

- Change Master User Password.

```
aws rds --profile sbg-psp-prod --region us-west-2 modify-db-instance --db-instance-identifier pspparmo-new --master-user-password xxxxxxxx
```

- Test connectivity to newly restored instance and change schema password

```
intuadmin/"xxxxx"@"pspparmo-new.cjls0bohfgpq.us-west-2.rds.amazonaws.com:2632/pspparmo'  
alter user pspadm identified by "xxxxxx";
```

- login to Monolith us-west GG hub

```
cd /u01/parallel-setup/
@cr_psp_prl_app_usr.sql
--password xxxxxxxx

@cr_psp_prl_read_usr.sql
--password xxxx
```

- create application user and read user

```
pspadm/"xxxxxxx"@'pspparmo-new.cjls0bohfgpq.us-west-2.rds.amazonaws.com:2632/pspparmo'

@installdb.sql PSP_PRL_APP PSPAPP_ROLE CRUD PSP_PRL_READ PSPREAD_ROLE

Expected Output:
=====
SQL> @installdb.sql PSP_PRL_APP PSPAPP_ROLE CRUD PSP_PRL_READ PSPREAD_ROLE

START TIME
-----
06/13/2022 08:06:48

USER Name
-----
PSPADM

=====
Step 1.Decide if it is a new install or an upgrade
=====

=====
Step 2.Create or update database objects (TABLES, INDEXES, FKS, PKS)
=====
SP2-0310: unable to open file "DBUpgrade.sql"
SP2-0310: unable to open file "DB_Update_Constraints.sql"

=====
Step 3. Populate tables and create procedures/triggers
=====

SP2-0310: unable to open file "DBPopulate.sql"

=====
Step 4.Create security objects (ROLES, SYNONIMS, GRANTS)
=====

=====
Step 5. Run Post installation Script
=====

SP2-0310: unable to open file "PostInstallDB.sql"

END TIME
-----
06/13/2022 08:06:53

END TIME
-----
06/13/2022 08:06:53
=====
```

- Lock application schema and user account on refreshed databaes

```
intuadmin/"xxxxx"@"pspparmo-new.cjls0bohfgpq.us-west-2.rds.amazonaws.com:2632/pspparmo'  
alter user pspadm account lock;  
alter user PSPAPP account lock;
```

- Validate user accounts are locked

```
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');
```

- Validate connectivity via application user to database

```
psp_prl_app/"xxxxx"@"pspparmo.cjls0bohfgpq.us-west-2.rds.amazonaws.com:2632/pspparmo'
```

- change awr snapshot period to every 15 mins and retention to 90 days.

```
BEGIN  
  DBMS_WORKLOAD_REPOSITORY.modify_snapshot_settings(  
    retention => 129600,      -- Minutes (= 90 Days).  
    interval  => 15);        -- Minutes.  
END;  
/
```

Post refresh steps for Monolith Oracle staging

Oracle Range & Oracle Hash Staging database

- Grant select privilege to PSP_PRL_APP user as per below on both Oracle Monolith Range and Oracle Hash Staging databases

```
grant select on V$TRANSACTION to PSP_PRL_APP;  
grant select on V$SESSION to PSP_PRL_APP;  
--check supplemental logging  
SELECT supplemental_log_data_min FROM v$database;  
--if not enabled  
To set up supplemental logging  
Run the following command to enable supplemental logging at the database level.  
exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD');  
  
Run the following command to enable primary key supplemental logging.  
exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD','PRIMARY KEY');  
  
(Optional) Enable key-level supplemental logging at the table level.  
  
ex:  
alter table table_name add supplemental log data (PRIMARY KEY) columns;
```

- run below on both

```

UPDATE PSPADM.PSP_SYSTEM_PARAMETER
SET SYSTEM_PARAMETER_VALUE='true'
WHERE SYSTEM_PARAMETER_CD = 'JPMC_ENABLE_ENCRYPTION';

UPDATE PSPADM.PSP_SYSTEM_PARAMETER
SET SYSTEM_PARAMETER_VALUE=NULL
WHERE SYSTEM_PARAMETER_CD='BANK_SFTP_ACH_HOST';

UPDATE PSPADM.PSP_SYSTEM_PARAMETER
SET SYSTEM_PARAMETER_VALUE='1234567890'
WHERE SYSTEM_PARAMETER_CD = 'JPMC_IMMEDIATE_ORIGIN';

UPDATE PSPADM.PSP_SYSTEM_PARAMETER
SET SYSTEM_PARAMETER_VALUE='1234567890'
WHERE SYSTEM_PARAMETER_CD = 'JPMC_IMMEDIATE_DESTINATION';

UPDATE PSPADM.PSP_SYSTEM_PARAMETER
SET SYSTEM_PARAMETER_VALUE='60000'
WHERE SYSTEM_PARAMETER_CD = 'JPMC_NACHA_FILE_UPLOAD_DELAY';

```

- As a Part of Lazy loading will run table scan and Index scan

```

scripts location: ggpspuwp01.sbg-ppsp-prod.a.intuit.com
cd /u01/mayank/fs
run scripts:
Eg: ./run_sql_pspdpdg.sh 1_tab_fs.sql--Enter Password

```

- create user for performance testing and dbzuser

```

CREATE USER Perf_test IDENTIFIED BY "XXX"
  DEFAULT TABLESPACE USERS
  TEMPORARY TABLESPACE TEMP
  PROFILE INDIVIDUAL_PROFILE;
GRANT PSPREAD_ROLE TO Perf_test;
GRANT CREATE SESSION TO Perf_test;
GRANT SELECT_CATALOG_ROLE TO PERF_TEST;
ALTER USER Perf_test DEFAULT ROLE ALL;
grant execute on rdsadmin.rdsadmin_util to perf_test;
GRANT "PSPAPP_ROLE" TO perf_test;
grant create any procedure to PERF_TEST;
grant execute on dbms_workload_repository to perf_test;
grant select any dictionary to perf_test;
grant execute on dbms_workload_repository to perf_test;
GRANT EXECUTE ON "PSPADM"."PK_GEMS_ACCOUNTS_RECEIVABLE" TO perf_test;
GRANT EXECUTE ON "PSPADM"."FN_GET_EDR_AMOUNT" TO perf_test;
GRANT EXECUTE ON "PSPADM"."FN_GET_LEDGER_BALANCE" TO perf_test;
GRANT DELETE ON "PSPADM"."PSP_PERF_SST" TO perf_test;
GRANT DELETE ON "PSPADM"."PSP_ADE_LAW_MAP" TO perf_test;
GRANT DELETE ON "PSPADM"."PSP_EMPLOYER_PREFERENCE" TO perf_test;
GRANT DELETE ON "PSPADM"."PSP_TAX_CREDITS_APPLICATION" TO perf_test;
GRANT DELETE ON "PSPADM"."PSP_FRAUD_CONTACT" TO perf_test;
GRANT DELETE ON "PSPADM"."PSP_FRAUD_COMPANY" TO perf_test;
GRANT DELETE ON "PSPADM"."PSP_FRAUD_ADDRESS" TO perf_test;
GRANT DELETE ON "PSPADM"."PSP_SQL_EXECUTION_LOG_ENTRY" TO perf_test;
GRANT DELETE ON "PSPADM"."PSP_PAY_ITEM" TO perf_test;
GRANT DELETE ON "PSPADM"."PSP_ENTITY_UPDATE" TO perf_test;
GRANT INSERT ON "PSPADM"."PSP_PERF_SST" TO perf_test;
GRANT INSERT ON "PSPADM"."PSP_ADE_LAW_MAP" TO perf_test;
GRANT INSERT ON "PSPADM"."PSP_EMPLOYER_PREFERENCE" TO perf_test;
GRANT INSERT ON "PSPADM"."PSP_TAX_CREDITS_APPLICATION" TO perf_test;
GRANT INSERT ON "PSPADM"."PSP_FRAUD_CONTACT" TO perf_test;
GRANT INSERT ON "PSPADM"."PSP_FRAUD_COMPANY" TO perf_test;
GRANT INSERT ON "PSPADM"."PSP_FRAUD_ADDRESS" TO perf_test;
GRANT INSERT ON "PSPADM"."PSP_SQL_EXECUTION_LOG_ENTRY" TO perf_test;
GRANT INSERT ON "PSPADM"."PSP_PAY_ITEM" TO perf_test;
GRANT INSERT ON "PSPADM"."PSP_ENTITY_UPDATE" TO perf_test;
GRANT SELECT ON "PSPADM"."SEQ_ATF_BATCH_ID_NBR" TO perf_test;

```

```

GRANT SELECT ON "PSPADM"."PSP_PERF_SST" TO perf_test;
GRANT SELECT ON "PSPADM"."PSP_ADE_LAW_MAP" TO perf_test;
GRANT SELECT ON "PSPADM"."PSP_EMPLOYER_PREFERENCE" TO perf_test;
GRANT SELECT ON "PSPADM"."PSP_TAX_CREDITS_APPLICATION" TO perf_test;
GRANT SELECT ON "PSPADM"."PSP_FRAUD_CONTACT" TO perf_test;
GRANT SELECT ON "PSPADM"."PSP_FRAUD_COMPANY" TO perf_test;
GRANT SELECT ON "PSPADM"."PSP_FRAUD_ADDRESS" TO perf_test;
GRANT SELECT ON "PSPADM"."PSP_SQL_EXECUTION_LOG_ENTRY" TO perf_test;
GRANT SELECT ON "PSPADM"."PSP_PAY_ITEM" TO perf_test;
GRANT SELECT ON "PSPADM"."PSP_ENTITY_UPDATE" TO perf_test;
GRANT UPDATE ON "PSPADM"."PSP_PERF_SST" TO perf_test;
GRANT UPDATE ON "PSPADM"."PSP_ADE_LAW_MAP" TO perf_test;
GRANT UPDATE ON "PSPADM"."PSP_EMPLOYER_PREFERENCE" TO perf_test;
GRANT UPDATE ON "PSPADM"."PSP_TAX_CREDITS_APPLICATION" TO perf_test;
GRANT UPDATE ON "PSPADM"."PSP_FRAUD_CONTACT" TO perf_test;
GRANT UPDATE ON "PSPADM"."PSP_FRAUD_COMPANY" TO perf_test;
GRANT UPDATE ON "PSPADM"."PSP_FRAUD_ADDRESS" TO perf_test;
GRANT UPDATE ON "PSPADM"."PSP_SQL_EXECUTION_LOG_ENTRY" TO perf_test;
GRANT UPDATE ON "PSPADM"."PSP_PAY_ITEM" TO perf_test;
GRANT UPDATE ON "PSPADM"."PSP_ENTITY_UPDATE" TO perf_test;
GRANT EXECUTE ON "PSPADM"."PK_GEMS_ACCOUNTS_RECEIVABLE" TO perf_test;
GRANT EXECUTE ON "PSPADM"."FN_GET_EDR_AMOUNT" TO perf_test;
GRANT EXECUTE ON "PSPADM"."FN_GET_LEDGER_BALANCE" TO perf_test;

CREATE TABLESPACE logminer_tbs DATAFILE SIZE 10G AUTOEXTEND ON NEXT 1G MAXSIZE UNLIMITED;
CREATE USER dbzuser IDENTIFIED BY "XXX" DEFAULT TABLESPACE LOGMINER_TBS QUOTA UNLIMITED ON LOGMINER_TBS;
GRANT CREATE SESSION TO dbzuser;
GRANT SET CONTAINER TO dbzuser;
GRANT SELECT ON V$DATABASE TO dbzuser;
GRANT FLASHBACK ANY TABLE TO dbzuser;
GRANT SELECT ANY TABLE TO dbzuser;
GRANT SELECT_CATALOG_ROLE TO dbzuser;
GRANT EXECUTE_CATALOG_ROLE TO dbzuser;
GRANT SELECT ANY TRANSACTION TO dbzuser;
GRANT SELECT ANY DICTIONARY TO dbzuser;
GRANT LOGMINING TO dbzuser;
GRANT CREATE TABLE TO dbzuser;
GRANT LOCK ANY TABLE TO dbzuser;
GRANT CREATE SEQUENCE TO dbzuser;
exec rdsadmin.rdsadmin_util.grant_sys_object('V_$LOG', 'DBZUSER', 'SELECT');
exec rdsadmin.rdsadmin_util.grant_sys_object('V_$LOG_HISTORY', 'DBZUSER', 'SELECT');
exec rdsadmin.rdsadmin_util.grant_sys_object('V_$LOGMNR_LOGS', 'DBZUSER', 'SELECT');
exec rdsadmin.rdsadmin_util.grant_sys_object('V_$LOGMNR_CONTENTS', 'DBZUSER', 'SELECT');
exec rdsadmin.rdsadmin_util.grant_sys_object('V_$LOGMNR_PARAMETERS', 'DBZUSER', 'SELECT');
exec rdsadmin.rdsadmin_util.grant_sys_object('V_$LOGFILE', 'DBZUSER', 'SELECT');
exec rdsadmin.rdsadmin_util.grant_sys_object('V_$ARCHIVED_LOG', 'DBZUSER', 'SELECT');
exec rdsadmin.rdsadmin_util.grant_sys_object('V_$ARCHIVE_DEST_STATUS', 'DBZUSER', 'SELECT');
exec rdsadmin.rdsadmin_util.grant_sys_object('V_$TRANSACTION', 'DBZUSER', 'SELECT');

```

Oracle Hash Staging

- Currently we have created global indexes on partitioned tables to support data validation using veridata. Drop these Global indexes on **Oracle Hash database Only (PSPPARHS)**
 - [temp_drop_gl_indexes_for_val.sql](#)
- As a Part of Lazy loading will run table scan and Index scan

```

scripts location: ggpspuwp01.sbg-psp-prod.a.intuit.com
/u01/mayank/fs/Hash-scan
run scripts:
Eg: ./run_sql_pspgardg.sh 1_tab_fs.sql--Enter Password

```

- create read only user,data lake team to test their jobs. **EBIETL_HASH_STG**

```
/u01/scripts
SQL> @cr_user_R0.sql
```

Refresh Staging Aurora Postgres Audit database

- Restore latest Prod snapshot to new instance psp-par-aud-new. Refer existing staging database (psp-par-aud) for configuration/parameter reference.

Restore snapshot

You are creating a new DB instance or DB cluster from a snapshot. The default VPC security group and parameter group are selected for the new DB instance or DB cluster, but you can change these settings.

DB instance settings

DB engine

Amazon Aurora PostgreSQL-Compatible Edition

Capacity type [Info](#)

☒ Provisioned

You provision and manage the server instance sizes.

Engine version [Info](#)

View the engine versions that support the following database features.

► [Show filters](#)

Available versions (4/4)

Aurora PostgreSQL (Compatible with PostgreSQL 14.3)

To see more versions, modify the capacity types. [Info](#)

Settings

DB snapshot ID

The identifier for the DB snapshot.

rds:psp-prod-ibob-2023-03-17-05-10

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

psp-par-aud-new

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Availability & durability

Multi-AZ deployment [Info](#)

☒ Create an Aurora Replica or Reader node in a different AZ (recommended for scaled availability)

Creates an Aurora Replica for fast failover and high availability.

☐ Don't create an Aurora Replica

Virtual private cloud (VPC) Info

Choose the VPC. The VPC defines the virtual networking environment for this DB cluster.

vpc-2 (vpc-0d8f1534aab54ac6) ▼

Only VPCs with a corresponding DB subnet group are listed.

ⓘ

After a database is created, you can't change its VPC.

DB subnet group Info

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB cluster can use in the VPC that you selected.

prod-clusterdb-vpc-2 ▼

Public access Info

- ☐ Yes
RDS assigns a public IP address to the cluster. Amazon EC2 instances and other resources outside of the VPC can connect to your cluster. Resources inside the VPC can also connect to the cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.
- ☒ No
RDS doesn't assign a public IP address to the cluster. Only Amazon EC2 instances and other resources inside the VPC can connect to your cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.

VPC security group (firewall) Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing
Choose existing VPC security groups

☐ Create new
Create new VPC security group

Existing VPC security groups

Choose one or more options ▼

parallel-clusterdb-vpc-2 ✕

bastionToDB-BastionToDBSecurityGroup-For-Parallel ✕

Availability Zone Info

No preference ▼

Additional configuration

Database port Info

TCP/IP port that the database will use for application connections.

6543

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class Info

- ☐ Serverless
- ☒ Memory optimized classes (includes r classes)
- ☐ Burstable classes (includes t classes)

db.r6g.16xlarge
64 vCPUs 512 GiB RAM Network: 19,000 Mbps ▼

☐ Include previous generation classes

Database authentication [Info](#)

Password authentication is always active for your database engine. You can also turn on additional authentication methods for your database below.

- ☐ **IAM database authentication**
Authenticates using IAM database authentication.
- ☐ **Kerberos authentication**
Authenticates using Kerberos authentication through an AWS Directory Service for Microsoft Active Directory.

Encryption

Encryption [Info](#)

- ☒ **Enable Encryption**
Choose to encrypt the copy of the source DB snapshot. Master key IDs and aliases appear in the list after they have been created using KMS. You cannot remove encryption from an encrypted DB snapshot.

AWS KMS key [Info](#)

psp-prod-ibob-cmk

Account

893547637742

KMS key ID

fa954df1-35f8-4f47-98c7-1f1a4d304560

▼ Additional configuration

Database options, backup turned on, backtrack turned off, CloudWatch Logs, maintenance, delete protection turned on

Database options

DB cluster parameter group [Info](#)

psp-shared-ibobpg14clusterf2200015-3z1tuohaihws

DB parameter group [Info](#)

psp-shared-ibobpg14instancea854b0ef-ey9i1itmbz2v

Option group [Info](#)

default:aurora-postgresql-14

Backup

- ☒ Copy tags to snapshots

Log exports

Select the log types to publish to Amazon CloudWatch Logs

- ☒ PostgreSQL log

IAM role

The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS service-linked role

 Ensure that general, slow query, and audit logs are turned on. Error logs are enabled by default. [Learn more](#)

Maintenance

Auto minor version upgrade [Info](#)

- ☐ **Enable auto minor version upgrade**
Enabling auto minor version upgrade will automatically upgrade to new minor versions as they are released. The automatic upgrades occur during the maintenance window for the database.

Deletion protection

- ☒ **Enable deletion protection**
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

Cancel

Restore DB cluster

- Connect to newly restored audit postgres db

```
ssh -o ServerAliveInterval=45 -i ~/.ssh/olympus_rsa_psp_prd -L 127.0.0.1:11999:psp-par-aud-new-cluster.cluster-cjls0bohfgppq.us-west-2.rds.amazonaws.com:6543 bastion-ppsp-prod-uw2 -A -f -N
```

```
psql -h 127.0.0.1 -p 11999 -U postgres -d postgres
```

- Ensure there is no connections from application user

```
SELECT datname as db,username as  
       username,client_addr as machine, count(*) FROM pg_stat_activity  
where username not in ('rdsadmin','postgres')  
group by db,username,machine;
```

```
db | username | machine | count  
----+-----+-----+-----  
(0 rows)
```

- Execute below in same sequence

```
alter database prodapgib rename to parapgib;  
  
--connect to staging database as below  
\c parapgib  
  
create user ibob_prl_pspapp password 'xxxxxxx';  
  
--if user already exists then execute below to allow login and change password  
alter user ibob_prl_pspapp login password 'xxxxxxx';  
  
grant connect on database parapgib to ibob_prl_pspapp;  
grant usage on schema ibobadm to ibob_prl_pspapp;  
grant all on all tables in schema ibobadm to ibob_prl_pspapp;  
grant all on all sequences in schema ibobadm to ibob_prl_pspapp;  
grant all on all functions in schema ibobadm to ibob_prl_pspapp;  
alter default privileges in schema ibobadm grant all on tables to ibob_prl_pspapp;  
alter default privileges in schema ibobadm grant all on sequences to ibob_prl_pspapp;  
alter default privileges in schema ibobadm grant all on functions to ibob_prl_pspapp;  
alter user ibob_prl_pspapp set search_path to ibobadm;
```

- Validate connectivity via application user to database from laptop

```
psql -h 127.0.0.1 -p 11999 -U ibob_prl_pspapp -d parapgib  
psql -h 127.0.0.1 -p 11999 -U postgres -d parapgib
```

- Lock application schema and user account on refreshed databaes

```
psql -h 127.0.0.1 -p 11999 -U postgres -d parapgib  
alter user ibobadm_owner with NOLOGIN;  
alter user ibob_prod_pspapp with NOLOGIN;  
alter user ibobadm_readonly with NOLOGIN;
```

- Check account status(rolcanlogin f)

```
select rolname,rolcanlogin  
from pg_roles  
where rolname in ('ibobadm_owner','ibob_prod_pspapp','ibobadm_readonly');
```

Post refresh steps for Monolith Aurora Postgres staging

- Drop replication slots

```
select * from pg_replication_slots;
select pg_drop_replication_slot(slot_name) from pg_replication_slots;
--Rename DB Name
alter database pspapg02 rename to pitparmo;
connect admin Host:
ggpspadm01.sbg-ppsp-prod.a.intuit.com

----Run Analyze for all tables
Location:cd /u01/refresh_scripts/postgres/analyze
run : ./analyze_pitparmo-new.sh

---Run vacuum for all Tables.
Location:cd /u01/refresh_scripts/postgres/vacuum/
Run: vacuum_pitparmo-new.sh
```

- Execute below in same sequence

```
\c pitparmo
alter user pspadm_owner login password 'XXXXXX';
grant create on database pitparmo to pspadm_owner;
grant usage, create on schema pspadm to pspadm_owner;
grant connect on database pitparmo to pspadm_readonly_role;
grant usage on schema pspadm to pspadm_readonly_role;
grant select on all tables in schema pspadm to pspadm_readonly_role;
grant select on pg_stat_statements to pspadm_readonly_role;
alter default privileges in schema pspadm grant select on tables to pspadm_readonly_role;
grant connect on database pitparmo to pspadm_readwrite_role;
grant usage on schema pspadm to pspadm_readwrite_role;
grant select on pg_stat_statements to pspadm_readwrite_role;

#login to pspadm_owner

\c pitparmo pspadm_owner
grant select on all tables in schema pspadm to pspadm_readonly_role;
alter default privileges in schema pspadm grant select on tables to pspadm_readonly_role;
-- grant select, DML permission to all the existing tables and views if any to the role
grant select, insert, update, delete on all tables in schema pspadm to pspadm_readwrite_role;
-- grant select, DML permission on all tables and views at later date to the role
alter default privileges in schema pspadm grant select, insert, update, delete on tables to
pspadm_readwrite_role;
-- grant select, DML permission to all the existing sequences if any to the role
grant select, usage on all sequences in schema pspadm to pspadm_readwrite_role;
-- grant select, DML permission to all the sequences created at later date to the role
alter default privileges in schema pspadm grant select, usage on sequences to pspadm_readwrite_role;
-- grant permission on functions, procedures
grant execute on all functions in schema pspadm to pspadm_readwrite_role;
grant execute on all procedures in schema pspadm to pspadm_readwrite_role;
-- grant permission on functions, procedures
alter default privileges in schema pspadm grant execute on functions to pspadm_readwrite_role;

\c pitparmo postgres

create user psp_prl_app password 'XXXX';
alter user psp_prl_app login password 'XXXX';
grant pspadm_readwrite_role to psp_prl_app;
alter user psp_prl_app set search_path to pspadm;
ALTER role psp_prl_app SET work_mem TO '16MB';
ALTER role psp_prl_app SET max_parallel_workers_per_gather TO 2;

create user perf_test with password 'XXXX';
alter user perf_test with password 'XXXX';
--grant read only role to users
grant pspadm_readwrite_role to perf_test;
```

```

grant usage, create on schema pspadm to perf_test;
--set search path for pspapp
alter user perf_test set search_path to pspadm;
grant select on pg_stat_statements to perf_test;

create user psp_prl_read with password 'XXXXX';
alter user psp_prl_read with password 'XXXXX';
--grant read only role
grant pspadm_readonly_role to psp_prl_read;
--set search path
alter user psp_prl_read set search_path to pspadm;
--set resource usage
ALTER role psp_prl_read SET work_mem TO '64MB';
ALTER role psp_prl_read SET max_parallel_workers_per_gather TO 4;

create user psprjf with password 'XXXXX';
Alter user psprjf with password 'XXXXXX';
--grant read only role
grant pspadm_readonly_role to psprjf;
--set search path
alter user psprjf set search_path to pspadm;
--set resource usage
ALTER role psprjf SET work_mem TO '64MB';
ALTER role psprjf SET max_parallel_workers_per_gather TO 4;

--# create read-write(OLAP) user/role with permission to login and grant needful permissions #--
create user pspbatch_rw_user with password 'XXXXX';
alter user pspbatch_rw_user login password 'XXXXX';
--grant read only role
grant pspadm_readwrite_role to pspbatch_rw_user;
--set search path
alter user pspbatch_rw_user set search_path to pspadm;
--set resource usage
ALTER role pspbatch_rw_user SET work_mem TO '64MB';
ALTER role pspbatch_rw_user SET max_parallel_workers_per_gather TO 4;

--# create read-only(OLAP) user/role with permission to login and grant needful permissions #--
create user pspbatch_ro_user with password 'XXXXX';
alter user pspbatch_ro_user login password 'XXXXX';
--grant read only role
grant pspadm_readonly_role to pspbatch_ro_user;
--set search path
alter user pspbatch_ro_user set search_path to pspadm;
--set resource usage
ALTER role pspbatch_ro_user SET work_mem TO '64MB';
ALTER role pspbatch_ro_user SET max_parallel_workers_per_gather TO 4;

```

- create **pspval** database and its related objects using [create-pspval-database.sql](#)

```

--connect to postgres staging

--run sql scripts from to create pspval database

```

- create tables under pspval schema in pspval database using [CreateCDCTables.sql](#)

```

--connect to pspval database in staging via pspval user
psql -h localhost -d pspval -U pspval -p 10543

--run sql scripts from above to create tables.
--truncate all tablestruncate table pspval.compare_info_cdc_event;
truncate table pspval.oracle_hash_cdc_event;
truncate table pspval.oracle_range_cdc_event;
truncate table pspval.postgres_hash_cdc_event;

```

- dbzuser access

```
grant create on database pitparmo to pspadm_owner;
grant usage, create on schema pspadm to pspadm_owner;
GRANT SET ON PARAMETER session_replication_role to pspadm_owner;
```

- Update Oracle Exit slack channel that staging db will not be available for 15 mins

Rename databases to actual staging db name

- Rename newly created RDS instance from **pspparmo** to **pspparmo-old** via aws console

RDS > Databases > Modify DB instance: pspparmo

Modify DB instance: pspparmo

Summary of modifications
You are about to submit the following modifications. Only values that will change are displayed. Carefully verify your changes and click Modify DB Instance.

Attribute	Current value	New value
DB instance identifier	pspparmo	pspparmo-old
Endpoint	pspparmo.cjs0bohfgpq.us-west-2.rds.amazonaws.com	pspparmo-old.cjs0bohfgpq.us-west-2.rds.amazonaws.com

Schedule modifications

When to apply modifications

☐ Apply during the next scheduled maintenance window
Current maintenance window: December 19, 2022 11:30 - 12:00 UTC+5.5

☒ **Apply immediately**
The modifications in this request and any pending modifications will be asynchronously applied as soon as possible, regardless of the maintenance window setting for this database instance.

Cancel Back **Modify DB instance**

- Rename newly created RDS instance from **psp-par-aud** to **psp-par-aud-old** via aws console
 - rename cluster

RDS > Databases > Modify DB cluster: psp-par-aud

Modify DB cluster: psp-par-aud

Summary of modifications
You are about to submit the following modifications. Only values that will change are displayed. Carefully verify your changes and click Modify cluster.

Attribute	Current value	New value
DB cluster identifier	psp-par-aud	psp-par-aud-old

Schedule modifications

When to apply modifications

☐ Apply during the next scheduled maintenance window
Current maintenance window: December 19, 2022 12:30 - 13:00 UTC+5.5

☒ **Apply immediately**
The modifications in this request and any pending modifications will be asynchronously applied as soon as possible, regardless of the maintenance window setting for this database cluster.

Cancel Back **Modify cluster**

- rename instance

RDS > Databases > Modify DB instance: psp-par-aud

Modify DB instance: psp-par-aud

Summary of modifications

You are about to submit the following modifications. Only values that will change are displayed. Carefully verify your changes and click Modify DB instance.

Attribute	Current value	New value
DB instance identifier	psp-par-aud	psp-par-aud-old
Endpoint	psp-par-aud.cjs0bohfgppq.us-west-2.rds.amazonaws.com	psp-par-aud-old.cjs0bohfgppq.us-west-2.rds.amazonaws.com

Schedule modifications

When to apply modifications

☐ Apply during the next scheduled maintenance window

Current maintenance window: December 19, 2022 13:01 - 13:31 UTC+5.5

☒ Apply immediately

The modifications in this request and any pending modifications will be asynchronously applied as soon as possible, regardless of the maintenance window setting for this database cluster.

Cancel

Back

Modify DB instance

RDS > Databases > Modify DB instance: psp-par-aud1

Modify DB instance: psp-par-aud1

Summary of modifications

You are about to submit the following modifications. Only values that will change are displayed. Carefully verify your changes and click Modify DB instance.

Attribute	Current value	New value
DB instance identifier	psp-par-aud1	psp-par-aud1-old
Endpoint	psp-par-aud1.cjs0bohfgppq.us-west-2.rds.amazonaws.com	psp-par-aud1-old.cjs0bohfgppq.us-west-2.rds.amazonaws.com

Schedule modifications

When to apply modifications

☐ Apply during the next scheduled maintenance window

Current maintenance window: December 19, 2022 13:01 - 13:31 UTC+5.5

☒ Apply immediately

The modifications in this request and any pending modifications will be asynchronously applied as soon as possible, regardless of the maintenance window setting for this database cluster.

Cancel

Back

Modify DB instance

- Rename newly created RDS instance from **pspparmo-new** to **pspparmo** via aws console

RDS > Databases > Modify DB instance: pspparmo-new

Modify DB instance: pspparmo-new

Summary of modifications

You are about to submit the following modifications. Only values that will change are displayed. Carefully verify your changes and click Modify DB Instance.

Attribute	Current value	New value
DB instance identifier	pspparmo-new	pspparmo
Endpoint	pspparmo-new.cjls0bohfgpq.us-west-2.rds.amazonaws.com	pspparmo.cjls0bohfgpq.us-west-2.rds.amazonaws.com

Schedule modifications

When to apply modifications

☐ Apply during the next scheduled maintenance window
Current maintenance window: December 19, 2022 11:30 - 12:00 UTC+5:5

☒ Apply immediately
The modifications in this request and any pending modifications will be asynchronously applied as soon as possible, regardless of the maintenance window setting for this database instance.

Cancel Back **Modify DB instance**

Security

VPC security groups

IKS-PRLRDS-VPC-Peering-IksRDSSecurityGroup
(sg-074b4dd587ff9746e)

✓ Active

- Rename newly created RDS instance from **psp-par-aud-new** to **psp-par-aud** via aws console
 - rename cluster

RDS > Databases > Modify DB cluster: psp-par-aud-new-cluster

Modify DB cluster: psp-par-aud-new-cluster

Summary of modifications

You are about to submit the following modifications. Only values that will change are displayed. Carefully verify your changes and click Modify cluster.

Attribute	Current value	New value
DB cluster identifier	psp-par-aud-new-cluster	psp-par-aud

Schedule modifications

When to apply modifications

☐ Apply during the next scheduled maintenance window

Current maintenance window: December 19, 2022 12:30 - 13:00 UTC+5.5

☒ Apply immediately

The modifications in this request and any pending modifications will be asynchronously applied as soon as possible, regardless of the maintenance window setting for this database cluster.

Cancel

Back

Modify cluster

Backup retention period [Info](#)

The number of days (1-35) for which automatic backups are kept.

7

▼

 days

Start time

05

▼

 :

00

▼

 UTC

Duration

0.5

▼

 hours

Log exports

Select the log types to publish to Amazon CloudWatch Logs

☒ PostgreSQL log

IAM role

The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS service-linked role

Security

VPC security groups

IKS-PRLRDS-VPC-Peering-IksRDSSecurityGroup
(sg-074b4dd587ff9746e)

 Active

- rename instance

RDS > Databases > Modify DB instance: psp-par-aud-new

Modify DB instance: psp-par-aud-new

Summary of modifications

You are about to submit the following modifications. Only values that will change are displayed. Carefully verify your changes and click Modify DB instance.

Attribute	Current value	New value
DB instance identifier	psp-par-aud-new	psp-par-aud
Endpoint	psp-par-aud-new.cjls0bohfgpq.us-west-2.rds.amazonaws.com	psp-par-aud.cjls0bohfgpq.us-west-2.rds.amazonaws.com

Schedule modifications

When to apply modifications

☐ Apply during the next scheduled maintenance window
Current maintenance window: December 17, 2022 15:14 - 15:44 UTC+5.5

☒ **Apply immediately**
The modifications in this request and any pending modifications will be asynchronously applied as soon as possible, regardless of the maintenance window setting for this database cluster.

- Finally it will be as below for monolith and audit staging databases

☐	DB identifier	▲	Role ▾	Engine ▾	Region & AZ ▾	Size ▾	Status ▾	CPU	Current activity	Maintenance ▾	VPC
☐	pspparmo		Instance	Oracle Enterprise Edition	us-west-2a	db.r5.16xlarge	Available	3.37%	217 Connections	available	vpc-0d8f1534aaa
☐	pspparmo-old		Source	Oracle Enterprise Edition	us-west-2b	db.r5.16xlarge	Available	0.16%	2 Connections	available	vpc-0d8f1534aaa

☐	DB identifier	▲	Role ▾	Engine ▾	Region & AZ ▾	Size ▾	Status ▾	CPU	Current activity	Maintenance ▾	VPC
☐	☐ psp-par-aud		Regional cluster	Aurora PostgreSQL	us-west-2	1 instance	Available	-		none	-
☐	☐ psp-par-aud		Writer instance	Aurora PostgreSQL	us-west-2b	db.r6g.16xlarge	Available	0.38%	167 Connections	none	vpc-0d8f1534aaab54
☐	☐ psp-par-aud-old		Regional cluster	Aurora PostgreSQL	us-west-2	2 instances	Available	-		available	-
☐	☐ psp-par-aud-old		Reader instance	Aurora PostgreSQL	us-west-2b	db.r6g.16xlarge	Available	0.32%	0.00 sessions	none	vpc-0d8f1534aaab54
☐	☐ psp-par-aud1-old		Writer instance	Aurora PostgreSQL	us-west-2a	db.r6g.16xlarge	Available	0.34%	0.00 sessions	available	vpc-0d8f1534aaab54

- only for postgres monolith (ppsp-stg-pitparmo) endpoint setup

- Before renaming cluster to old drop custom endpoints existing cluster **ppsp-stg-pitparmo** .

```
select machine, count(1)
--login console connect ppsp-stg-pitparmo
Delete below custom endpoints.
```

```
ppsp-stg-datalake.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com
ppsp-stg-rjf.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com
ppsp-stg-rpt.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com
ppsp-stg-vmp.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com
```

- create same custom endpoints under new cluster **ppsp-stg-pitparmo-new**.

- rename existing **psp-stg-pitparmo** to **psp-stg-pitparmo-old**.
- rename **ppsp-stg-pitparmo-new** to **ppsp-stg-pitparmo**.

Sanity Check

- Check connections in **pspparmo** (should have connections) and **pspparmo-old** (should have no connections)

```
select machine, count(1)
from gv$session
group by rollup(machine)
order by 1;
```

- Check connections in **pspparmo** (should have connections) and **pspparmo-old** (should have no connections)

```
set lines 300
col username for a30
col machine for a70;
select username,machine,count(*) from gv$session where username is not NULL and username not in
('RDSADMIN','INTUADMIN','SYS','SYSTEM') and type = 'USER' group by service_name,username,machine order by
username,machine;
```

- Check connections **psp-par-aud** (should have connections) and **psp-par-aud-old** (should have no connections)

```
SELECT datname as db,username as
  username,client_addr as machine, count(*) FROM pg_stat_activity
where username not in ('rdsadmin','postgres')
group by db,username,machine;
```

- If you are able to see connections on **pspparmo-old** and **psp-par-aud-old** . Reboot both the instances from console and check again connections.
- Enable Performance Insights for RDS (**free tier**) (7 days retention)
- Re-run Below scripts after rename

```
grant select on V$TRANSACTION to PSP_PRL_APP;
grant select on V$SESSION to PSP_PRL_APP;
```

Cleanup old staging databases

- **Drop old databases after 1 day.**